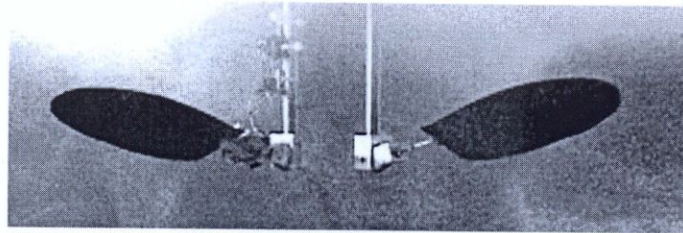


INTRODUCTION TO FLUID MECHANICS

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A paper published in Scientific American in 2001 by Prof. Michael Dickinson describes an experimental facility designed to determine the lift L generated by a fly suspended at a fixed point by the flapping motion of its wings of length L at a frequency f . The facility includes a large container filled with mineral oil. A pair of wings, geometrically similar to those of the fly, is placed within the container. Programmable electrical engines move the wings, reproducing the wing motion of the fly at varying frequencies. To mimic the characteristic fly dynamics, the wings in the experiment are made out of a very light material, so that their weight and inertia are entirely negligible. To guide the experiment design and analyze the results follow the steps indicated below.



1. Use dimensional analysis to express the lift force F as a function of the minimum number of dimensionless parameters.

$$F = F(f, L, \rho, \mu)$$

2. Assuming the following characteristic values for the fly

$$\frac{F}{\rho L^4 f^2} = \pi \left(\frac{\rho L^2 f}{\mu} \right)$$

- Flapping frequency: $f = 200$ Hz

- Wing length: $L = 2.5$ mm

- Air density: $\rho = 1.2$ kg/m³

- Air viscosity: $\mu = 1.8 \times 10^{-5}$ kg/m·s

$$\frac{\rho}{\mu} L^2 f = \frac{\rho_e}{\mu_e} L_e^2 f_e$$

along with the nominal characteristics of the experiment

$$\mu_e = \mu \frac{\rho}{\rho_e} \left(\frac{L_e}{L} \right)^2 \frac{f_e}{f} =$$

- Flapping frequency: $f_e = 0.2$ Hz

- Wing length: $L_e = 250$ mm

- Mineral oil density: $\rho_e = 800$ kg/m³,

$$= 1.8 \cdot 10^{-5} \frac{\text{kg}}{\text{m} \cdot \text{s}} \frac{800}{1.2} \left(\frac{250}{2.5} \right)^2 \frac{0.2}{200} = 0.12 \frac{\text{kg}}{\text{m} \cdot \text{s}}$$

determine the mineral-oil viscosity μ_e needed to guarantee that the experiment is a physically similar representation of the fly dynamics.

3. The experimental facility including the model with $L_e = 250$ mm and the mineral oil selected above is used to investigate the variation of the lift with the flapping frequency. To that end, the wings are moved at different frequencies f_e and the corresponding forces F_e are measured, yielding the data points shown in the table below. Generate a curve for the variation of the force F with the flapping frequency f for the fly of wing length $L = 2.5$ mm.

f_e (Hz)	F_e (N)
0.14	1.33
0.15	3.33
0.16	4.80
0.17	7.13
0.18	8.00
0.19	6.60
0.20	5.07
0.21	4.53

$$\frac{\rho L^2 f}{\mu} = \frac{\rho_e L_e^2 f_e}{\mu_e} \Rightarrow f = \frac{\rho_e}{\rho} \left(\frac{L_e}{L} \right)^2 \frac{\mu}{\mu_e} f_e$$

$$\frac{F}{\rho L^4 f^2} = \frac{F_e}{\rho_e L_e^4 f_e^2} \Rightarrow F = \frac{\rho}{\rho_e} \left(\frac{L_e}{L} \right)^4 \left(\frac{f}{f_e} \right)^2 F_e = \frac{\rho}{\rho_e} \left(\frac{\mu}{\mu_e} \right)^2 F_e$$

f (Hz)	F (N)
140	0.2×10^{-4}
150	0.5×10^{-4}
160	0.72×10^{-4}
170	1.07×10^{-4}
180	1.2×10^{-4}
190	1×10^{-4}
200	0.76×10^{-4}
210	0.68×10^{-4}